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Lab 2

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02.1: TCP, HTTP

1. TCP #1: Sockets
    - **Take a screenshot of the prompt and the command that ChatGPT generates**

What is a single Linux command using `ss` or `netstat` that lists all TCP sockets in the LISTEN state on IPv4 addresses and shows the program associated with each socket? Include the necessary command-line flags and use `sudo`.

Use the following Linux command:

Bash

```
sudo ss -4 -t -l -n -p
```

Explanation of the flags:

- `sudo` : Runs the command with administrator privileges.
- `ss` : Displays socket information.
- `-4` : Shows only IPv4 sockets.
- `-t` : Shows only TCP sockets.
- `-l` : Shows only sockets in the LISTEN state.
- `-n` : Displays numerical IP addresses and port numbers.
- `-p` : Shows the program name and process ID (PID) associated with each socket.

Result: This command lists all TCP sockets listening on IPv4

- **Run the command using sudo and take a screenshot of the output to include in your lab notebook.**

```
evelynng@course-vm:~$ sudo ss -4 -t -l -n -p
State     Recv-Q    Send-Q        Local Address:Port          Peer Address:Port
Process
LISTEN    0         128                 0.0.0.0:22                 0.0.0.0:*
 users:(("sshd",pid=841,fd=3))
LISTEN    0         4096          127.0.0.53%lo:53                 0.0.0.0:*
 users:(("systemd-resolve",pid=322,fd=14))
LISTEN    0         128                 0.0.0.0:20202              0.0.0.0:*
 users:(("fluent-bit",pid=857,fd=160))
LISTEN    0         4096             127.0.0.1:39679              0.0.0.0:*
 users:(("containerd",pid=414,fd=14))
evelynng@course-vm:~$ odinID: evelynng
```

- **List a service that can be contacted from any interface on the machine. List a service that can only be contacted by local processes**
    - Service can be contacted from any interface: sshd port 22
    - Service can be contacted by local processes: systemd-resolve (Port 53)
- **Run the command again, but do not use sudo as this is a machine managed by CAT. Include a screenshot of the output.**

```
evelynng@ada:~$ ss -4 -t -l -n -p
State    Recv-Q   Send-Q     Local Address:Port      Peer Address:Port   Process
LISTEN   0        4096       127.0.0.53%lo:53            0.0.0.0:*
LISTEN   0        4096          127.0.0.54:53            0.0.0.0:*
LISTEN   0        4096             0.0.0.0:22            0.0.0.0:*
LISTEN   0        100            127.0.0.1:25            0.0.0.0:*
LISTEN   0        4096           127.0.0.1:631           0.0.0.0:*
LISTEN   0        128            127.0.0.1:6100          0.0.0.0:*
LISTEN   0        128            127.0.0.1:6101          0.0.0.0:*
LISTEN   0        128            127.0.0.1:6102          0.0.0.0:*
LISTEN   0        128            127.0.0.1:6103          0.0.0.0:*
evelynng@ada:~$ odinID: evelynng
```

- **List the services that this machine provides for external access:**
    - only one ssh port 22

2. **lsof**
    - **Take a screenshot of the prompt and the command that ChatGPT generates**

repeat the previous task, but for lsof command

Use the following Linux command:

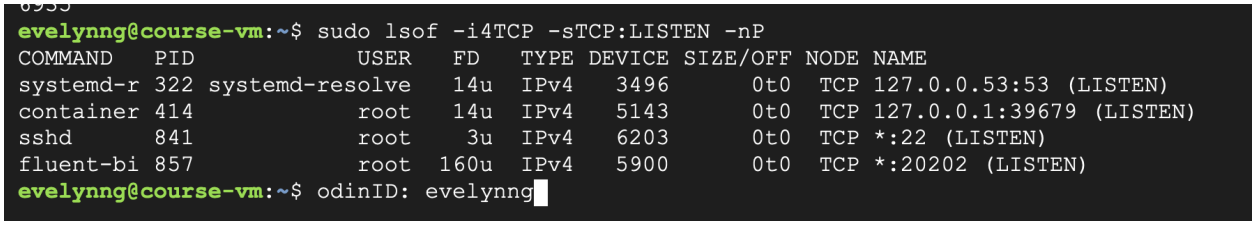
```bash
sudo lsof -nP -i4TCP -sTCP:LISTEN
```

Explanation of the flags:

- `sudo` : Runs the command with administrator privileges to display information about all processes.
- `lsof` : Lists open files, including network sockets.
- `-n` : Prevents DNS hostname resolution, displaying numerical IP addresses.
- `-P` : Displays numerical port numbers instead of service names.
- `-i4TCP` : Shows only IPv4 TCP sockets.
- `-sTCP:LISTEN` : Filters results to TCP sockets in the LISTEN state.

Result: This command lists all TCP sockets listening on IPv4 addresses, including the associated program name (`COMMAND`), process ID (`PID`), IP address, and port number.

- **Run the command using sudo and take a screenshot of the output to include in your lab notebook.**

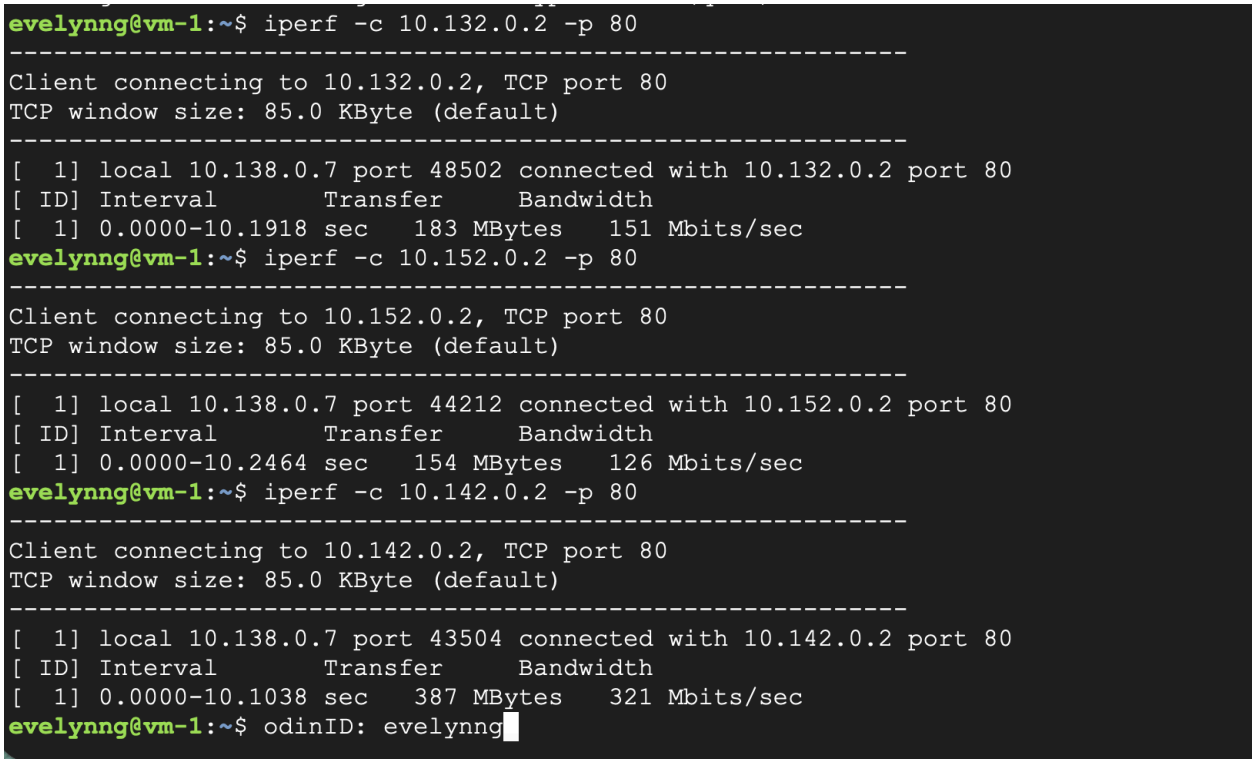


3. TCP #2: Throughput

- No questions

4. iperf

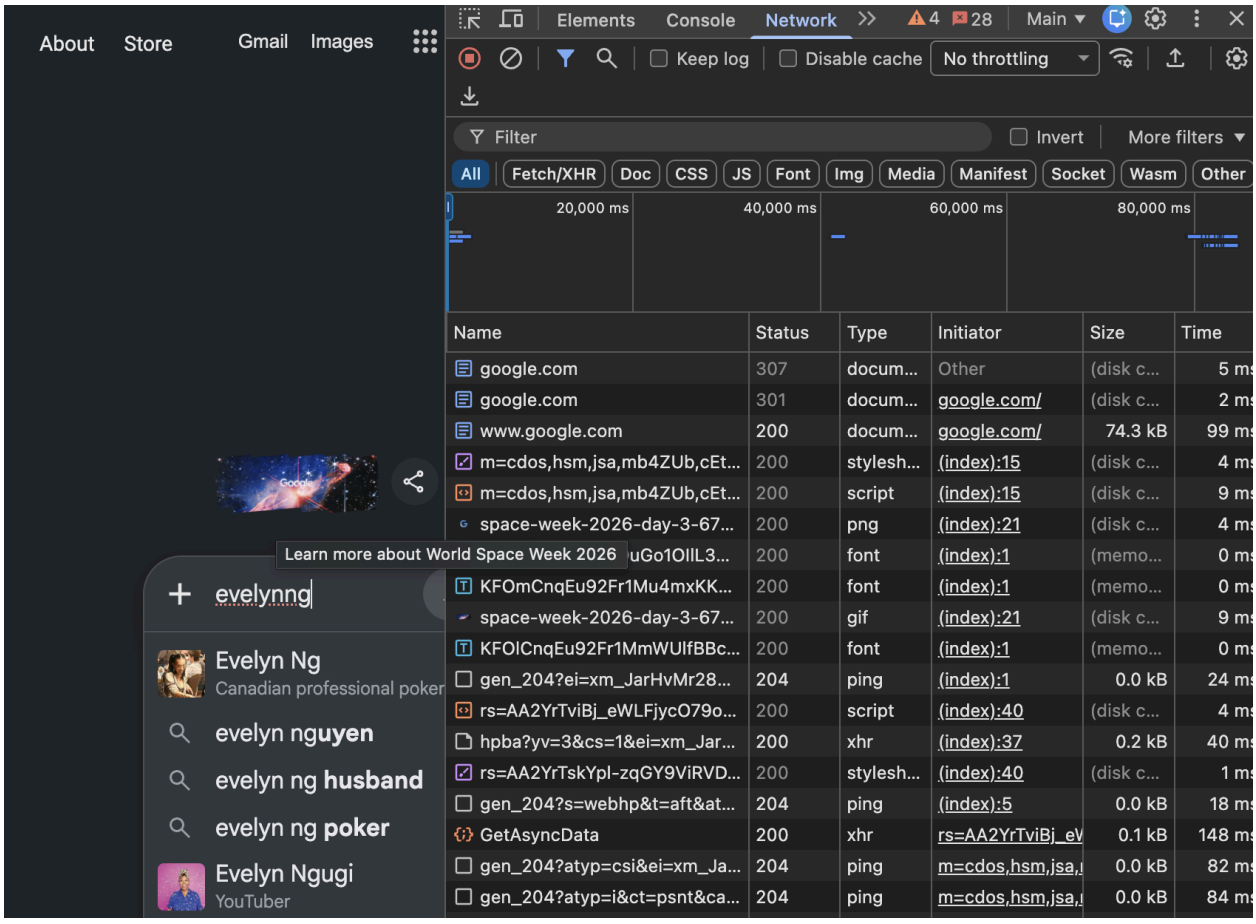
- **Show a screenshot of the measured bandwidth available between your us-west1-b VM and each of the other Compute Engine VMs. Explain the relative differences (or lack thereof) in your results.**



- These differences may be caused by geographical distance, network latency, and network conditions. Greater distances can increase latency, which may reduce TCP throughput.

5. HTTP #3: Requests

- **Take a screenshot of the initial requests for your lab notebook**



- **What is the URL being requested?**

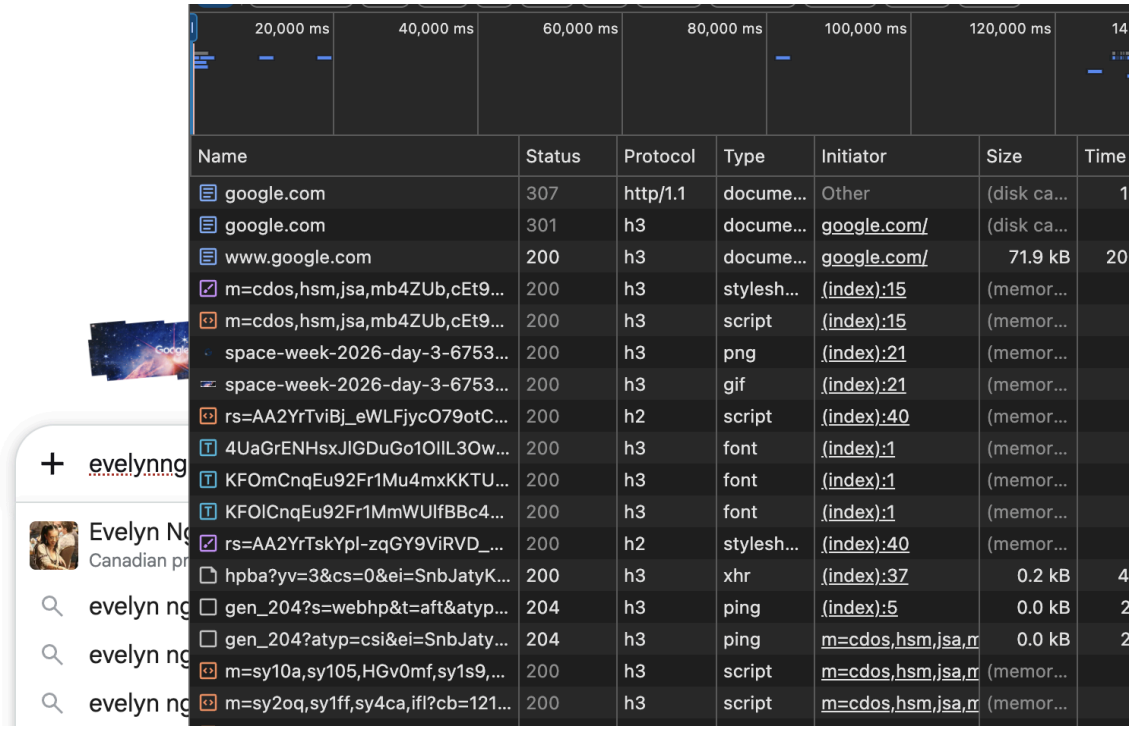
Request	URL	HTTP status

1	http://google.com/	307 Internal Redirect
2	https://google.com/	301 Moved Permanently
3	https://www.google.com/	200 OK

- **Explain the HTTP status code that is returned and what the code indicates**

- **307 Internal Redirect:** The browser internally redirects the original request from http://google.com/ to HTTPS. This is a temporary redirect performed by Chrome rather than an HTTP response from Google's server.
- **301 Moved Permanently:** Google's server indicates that https://google.com/ has permanently moved to https://www.google.com/. The browser follows the redirect to the new URL.
- **200 OK:** The request to https://www.google.com/ was successful, and the server returned the requested webpage.

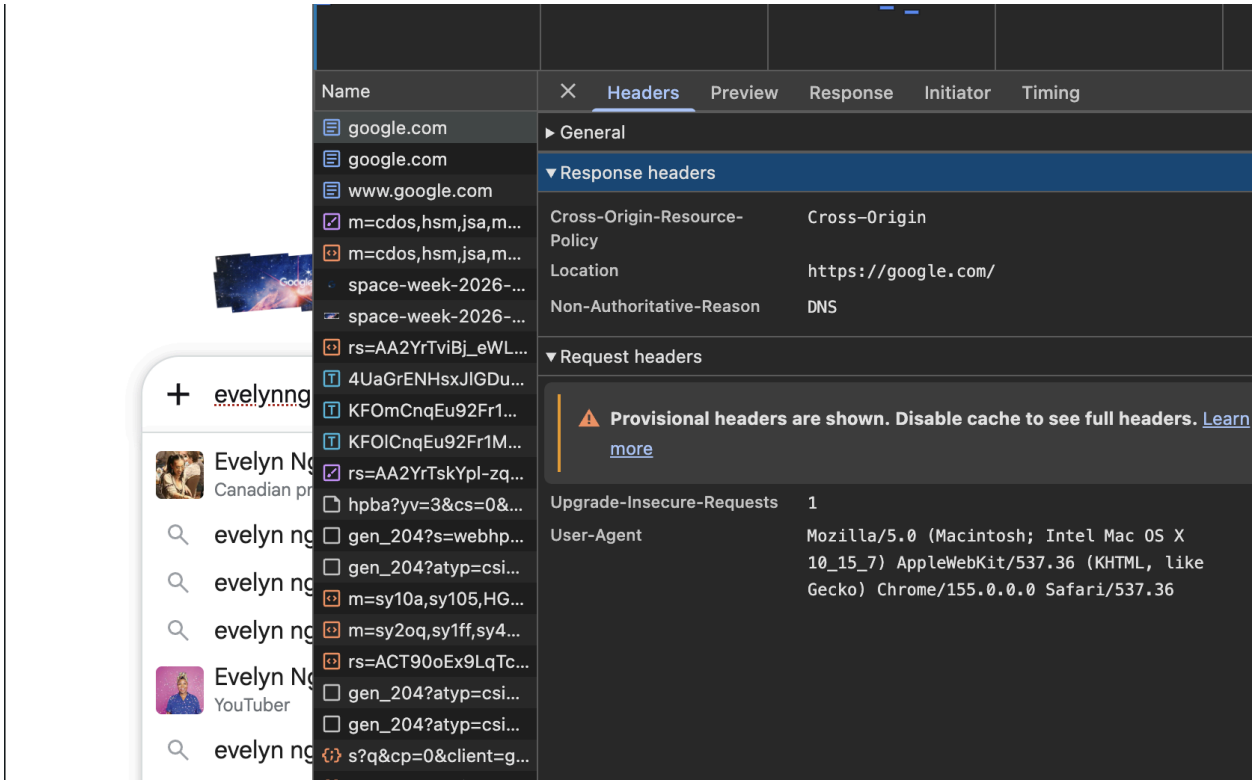
- **Take a screenshot indicating the version of the HTTP protocol that is used for each request. (Hint: look at the response status line and alt-svc: HTTP response headers indicating HTTP/2 or HTTP/3).**



- 1

  http://google.com/ (307)

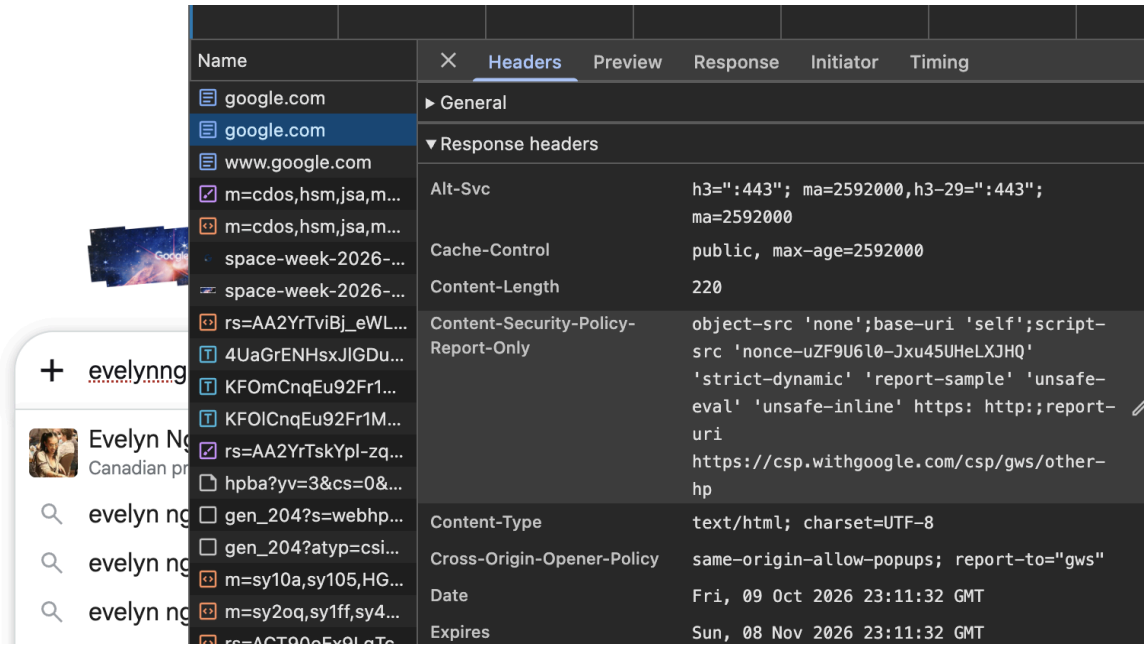
  Location: https://google.com/



- 2

  https://google.com/ (301)

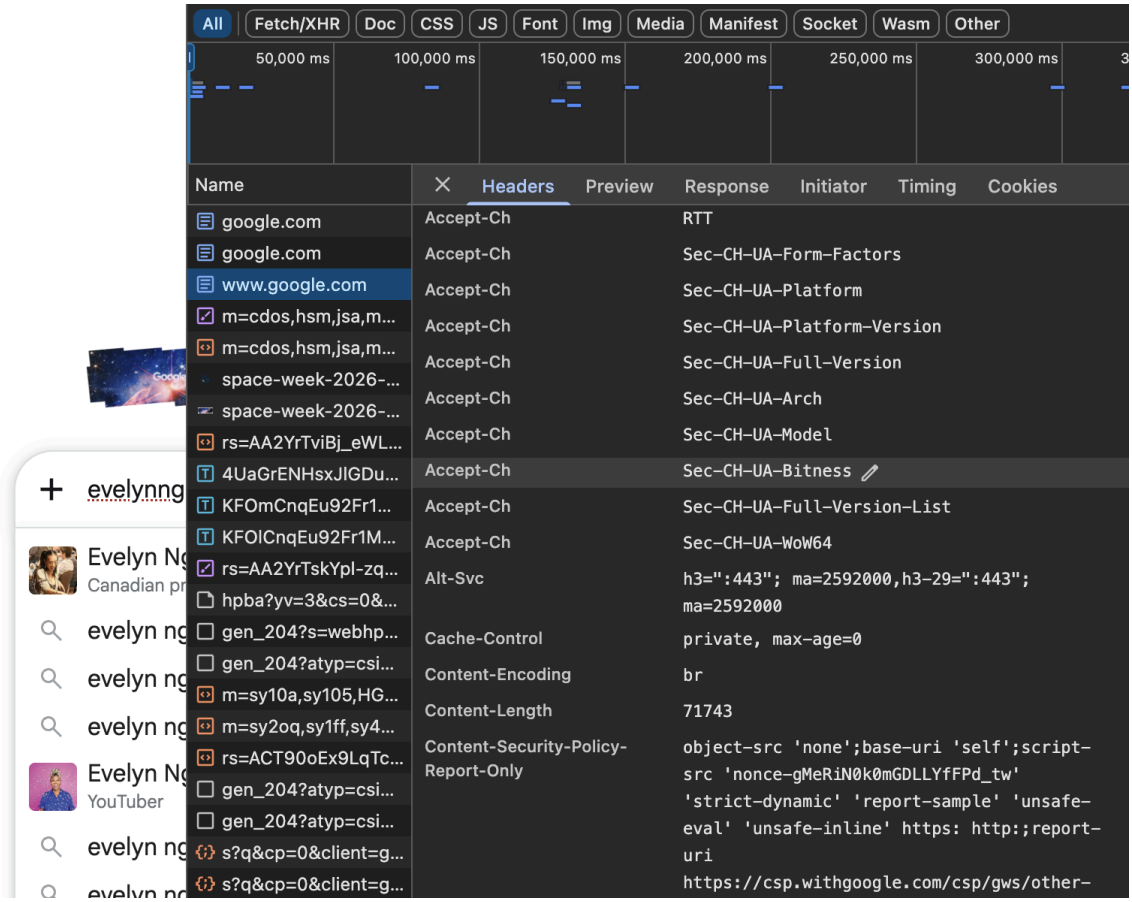
  Alt-Svc: h3=":443"



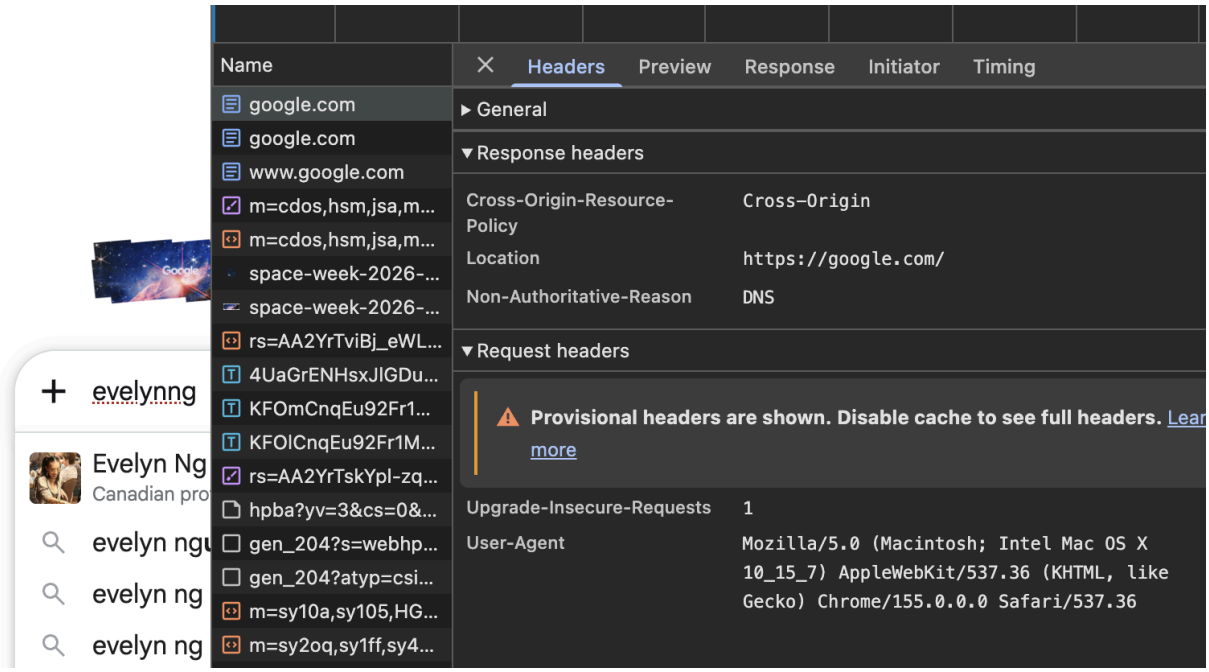
- 3

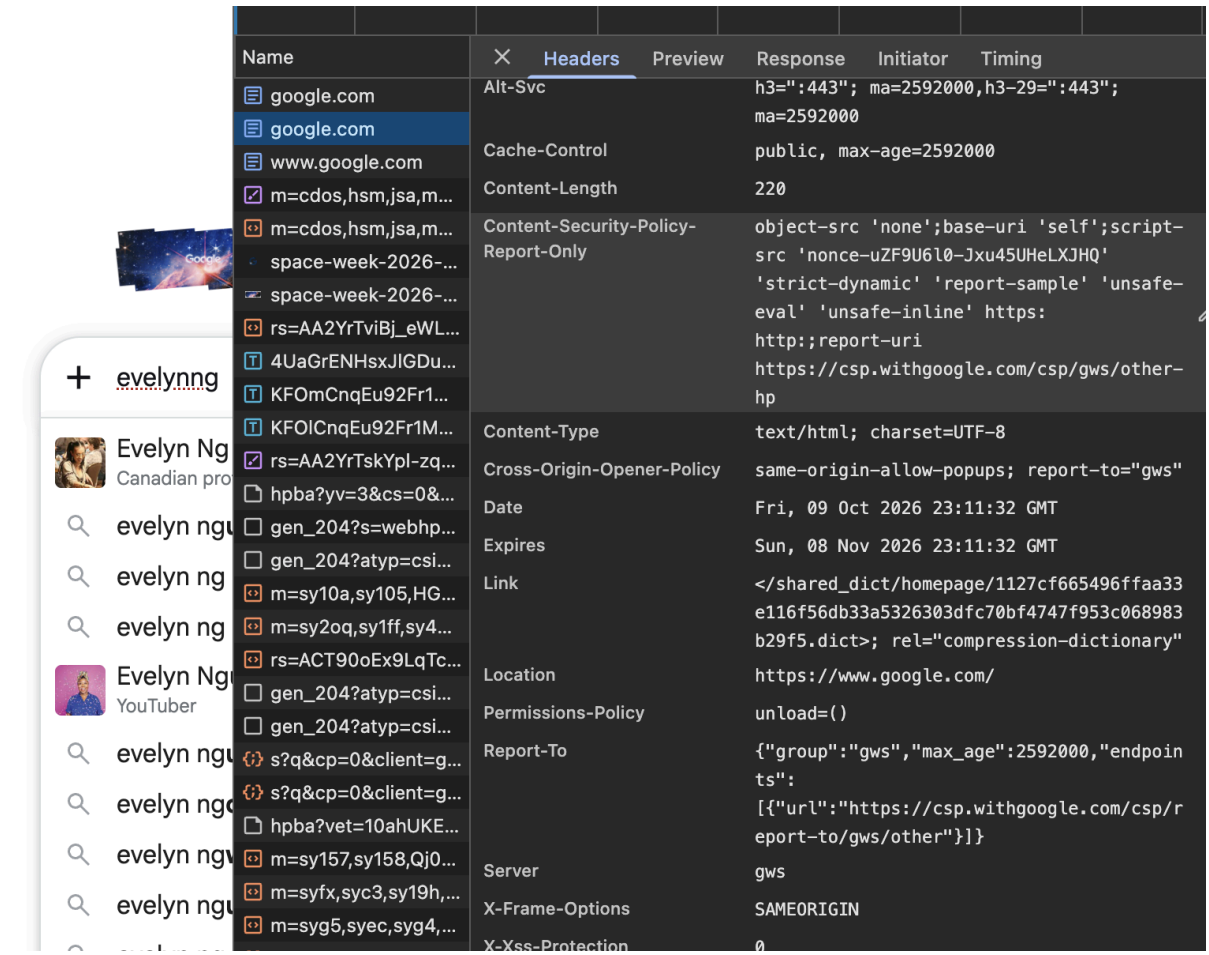
  https://www.google.com/ (200)

  Alt-Svc: h3=":

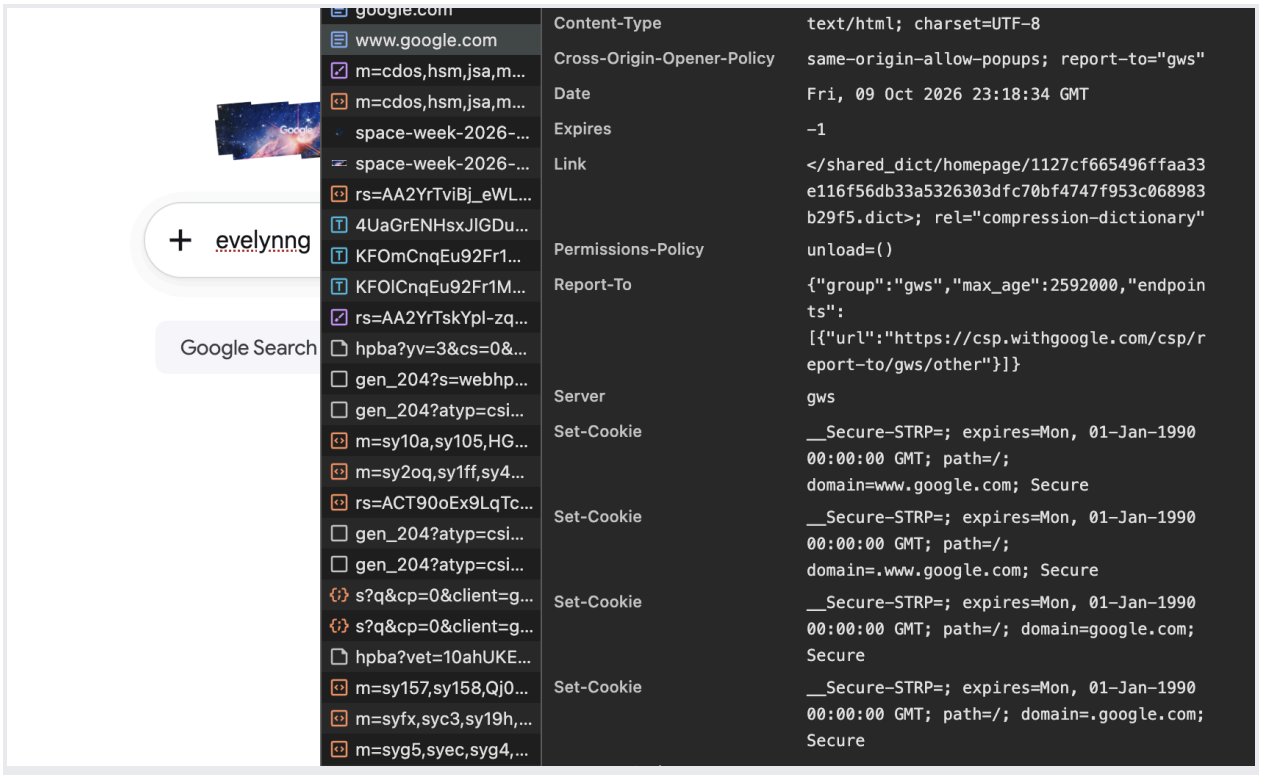

All
Fetch/XHR
Doc
CSS
JS
Font
Img
Media
Manifest
Socket
Wasm
Other
50,000 ms
100,000 ms
150,000 ms
200,000 ms
250,000 ms
300,000 ms
Name
Headers
Preview
Response
Initiator
Timing
Cookies
google.com
google.com
www.google.com
m=cdos,hsm,jsa,m...
m=cdos,hsm,jsa,m...
space-week-2026-...
space-week-2026-...
rs=AA2YrTviBj_eWL...
4UaGrENHsxJlGDu...
KFOmCnqEu92Fr1...
KFOlCnqEu92Fr1M...
rs=AA2YrTskYpI-zq...
hpba?yv=3&cs=0&...
gen_204?s=webhp...
gen_204?atyp=csi...
m=sy10a,sy105,HG...
m=sy2oq,sy1ff,sy4...
rs=ACT90oEx9LqTc...
gen_204?atyp=csi...
gen_204?atyp=csi...
s?q&cp=0&client=g...
s?q&cp=0&client=g...
Accept-Ch RTT
Accept-Ch Sec-CH-UA-Form-Factors
Accept-Ch Sec-CH-UA-Platform
Accept-Ch Sec-CH-UA-Platform-Version
Accept-Ch Sec-CH-UA-Full-Version
Accept-Ch Sec-CH-UA-Arch
Accept-Ch Sec-CH-UA-Model
Accept-Ch Sec-CH-UA-Bitness
Accept-Ch Sec-CH-UA-Full-Version-List
Accept-Ch Sec-CH-UA-WoW64
Alt-Svc h3=":443"; ma=2592000,h3-29=":443"; ma=2592000
Cache-Control private, max-age=0
Content-Encoding br
Content-Length 71743
Content-Security-Policy-Report-Only object-src 'none';base-uri 'self';script-src 'nonce-gMeRiN0k0mGDLLYfFPd_tw' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other-
evelynng
Evelyn Ng
Canadian pr
evelyn ng
evelyn ng
evelyn ng
Evelyn Ng
YouTuber
evelyn ng

- **Show the URLs the browser is redirected to via this header”**

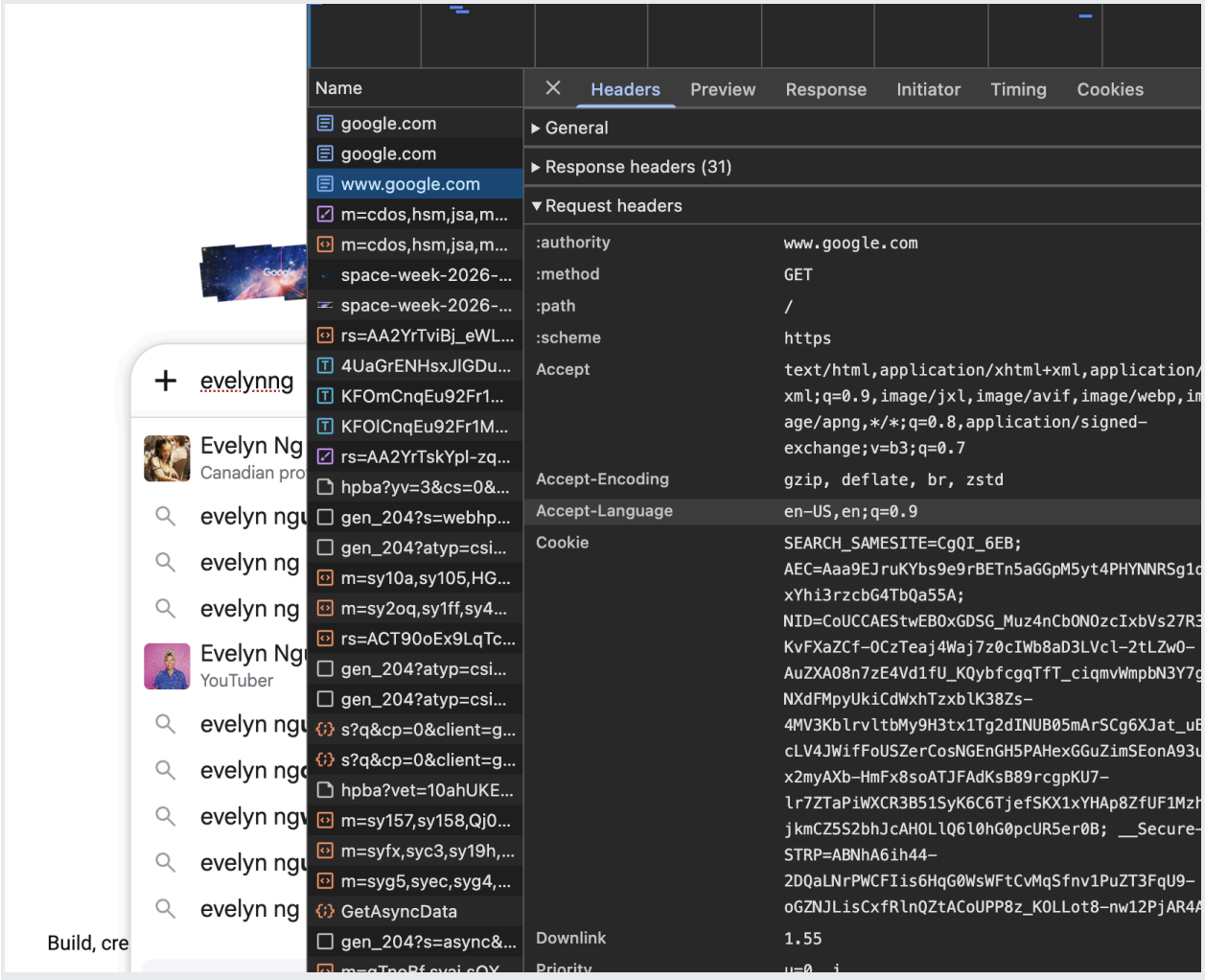




- **Take a screenshot of when cookies are set via Set-Cookie:**

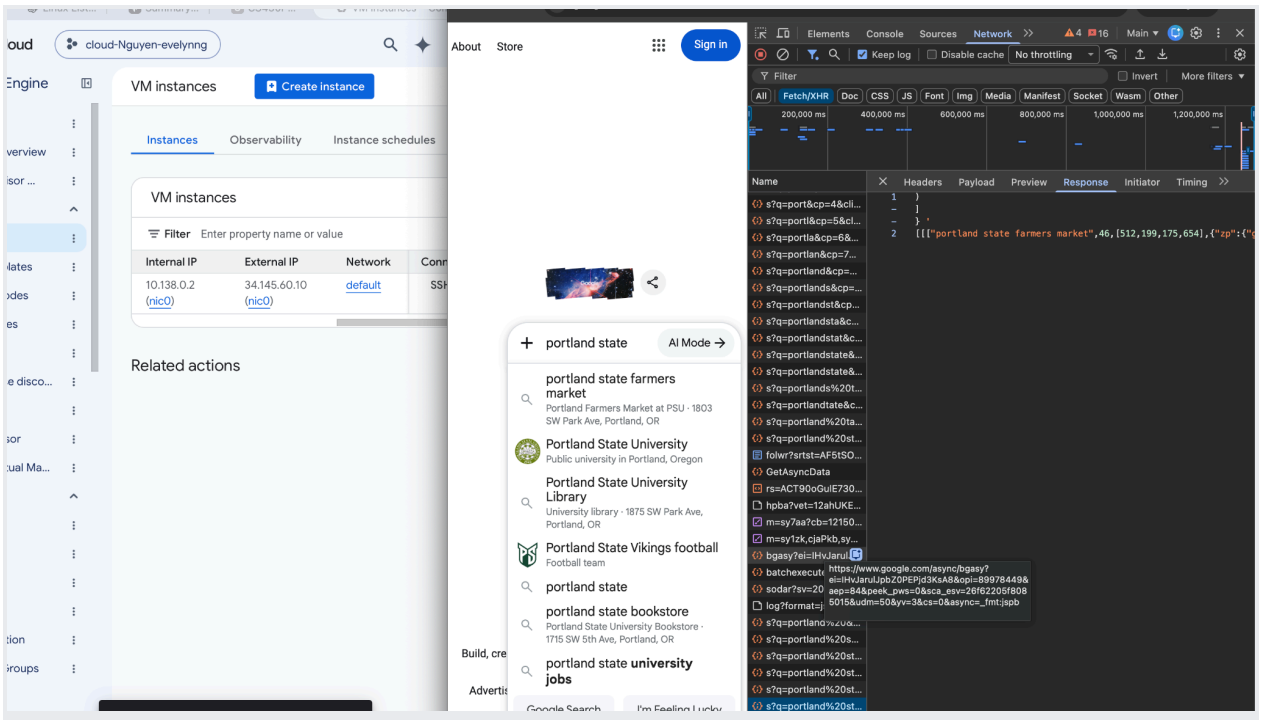


- **Take a screenshot of when cookies are attached via Cookie:**



6. Asynchronous http request

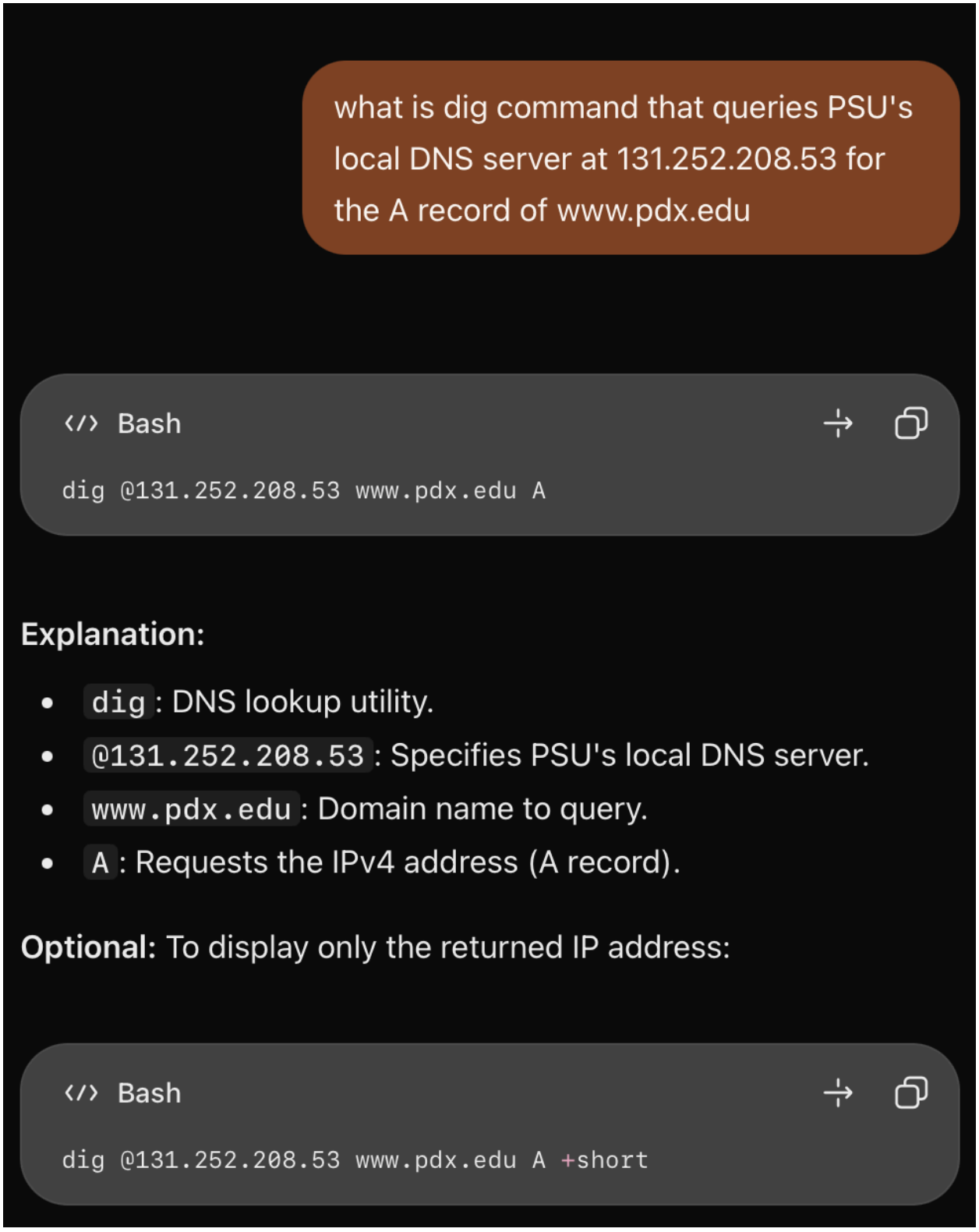
- **Show the requests and responses in the listing. Click on the last request sent, then click on the response to see that its payload has returned the data that is then rendered on the search page similar to what is shown below for "rabbid"**



02.2: DNS, Recap

1. DNS reconnaissance #1 (dig)

- **Take a screenshot of the prompt and the dig command produced.**



- **Take a screenshot of the records returned for your lab notebook**

```
evelynng@ada:~$ dig @131.252.208.53 www.pdx.edu A

; <<>> DiG 9.20.24-1ubuntu0.3-Ubuntu <<>> @131.252.208.53 www.pdx.edu A
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 33783
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: da94432db5ed8567010000006acc54c35634f82cb09c74d0 (good)
;; QUESTION SECTION:
;www.pdx.edu.                   IN      A

;; ANSWER SECTION:
www.pdx.edu.            60      IN      A       18.161.6.120
www.pdx.edu.            60      IN      A       18.161.6.96
www.pdx.edu.            60      IN      A       18.161.6.84
www.pdx.edu.            60      IN      A       18.161.6.112

;; Query time: 26 msec
;; SERVER: 131.252.208.53#53(131.252.208.53) (UDP)
;; WHEN: Sun Oct 11 20:32:19 PDT 2026
;; MSG SIZE  rcvd: 132
```

```
evelynng@ada:~$ dig @131.252.208.53 pdx.edu. MX

; <<>> DiG 9.20.24-1ubuntu0.3-Ubuntu <<>> @131.252.208.53 pdx.edu. MX
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 30288
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 8e1d5a0466b57415010000006acc57a93653069d36249eea (good)
;; QUESTION SECTION:
;pdx.edu.                       IN      MX

;; ANSWER SECTION:
pdx.edu.                62570   IN      MX      1 aspmx.l.google.com.
pdx.edu.                62570   IN      MX      10 alt3.aspmx.l.google.com.
pdx.edu.                62570   IN      MX      5 alt1.aspmx.l.google.com.
pdx.edu.                62570   IN      MX      5 alt2.aspmx.l.google.com.
pdx.edu.                62570   IN      MX      10 alt4.aspmx.l.google.com.

;; Query time: 1 msec
;; SERVER: 131.252.208.53#53(131.252.208.53) (UDP)
;; WHEN: Sun Oct 11 20:44:41 PDT 2026
;; MSG SIZE  rcvd: 182
```

- **What cloud provider hosts the web site for www.pdx.edu?**
  - Amazon.com Inc.
- **What cloud provider handles mail for pdx.edu?**
  - google
- **Take a screenshot of the results for both records for your lab notebook.**

```
evelynng@ada:~$ dig @ns-cloud-e2.googledomains.com mashimaro.cs.pdx.edu. NS +norecu
rse

; <<>> DiG 9.20.24-1ubuntu0.3-Ubuntu <<>> @ns-cloud-e2.googledomains.com mashimaro.
cs.pdx.edu. NS +norecurse
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 63076
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 3, ADDITIONAL: 7

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
; COOKIE: 5d40871f188b676d0169727a2f927637aed7d0f9d17c382392fdfacca7 (good)
;; QUESTION SECTION:
;mashimaro.cs.pdx.edu.          IN      NS

;; AUTHORITY SECTION:
cs.pdx.edu.             14400   IN      NS      adns3.cat.pdx.edu.
cs.pdx.edu.             14400   IN      NS      adns2.cat.pdx.edu.
cs.pdx.edu.             14400   IN      NS      adns1.cat.pdx.edu.

;; ADDITIONAL SECTION:
adns3.cat.pdx.edu.      14400   IN      A       52.11.85.139
adns3.cat.pdx.edu.      14400   IN      AAAA    2600:1f13:d7a:ef00:d17c:c07a:9bb8:3
48e
adns1.cat.pdx.edu.      14400   IN      A       131.252.208.38
adns1.cat.pdx.edu.      14400   IN      AAAA    2610:10:20:208::38
adns2.cat.pdx.edu.      14400   IN      A       35.83.149.241
adns2.cat.pdx.edu.      14400   IN      AAAA    2600:1f13:d7a:ef00:e934:6a5c:f7c4:3
e64

;; Query time: 113 msec
;; SERVER: 216.239.34.110#53(ns-cloud-e2.googledomains.com) (UDP)
;; WHEN: Sun Oct 11 23:10:35 PDT 2026
;; MSG SIZE  rcvd: 278
```

```
evelynng@ada:~$ dig @adns2.cat.pdx.edu mashimaro.cs.pdx.edu. A

; <<>> DiG 9.20.24-1ubuntu0.3-Ubuntu <<>> @adns2.cat.pdx.edu mashimaro.cs.pdx.edu.
A
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 63486
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 1a83f01f4de1cecf010000006acc7a5c9a0180503c0d1608 (good)
;; QUESTION SECTION:
;mashimaro.cs.pdx.edu.          IN      A

;; ANSWER SECTION:
mashimaro.cs.pdx.edu.   14400   IN      A       131.252.220.66

;; Query time: 6 msec
;; SERVER: 35.83.149.241#53(adns2.cat.pdx.edu) (UDP)
;; WHEN: Sun Oct 11 23:12:44 PDT 2026
;; MSG SIZE  rcvd: 93
```

2. Iterative DNS lookups

- **List all of the iterative dig commands performed for the lookup**

dig
dig @192.5.5.241 com. NS +norecurse +tcp
dig @a.gtld-servers.net google.com. NS +norecurse +tcp
dig @ns1.google.com console.cloud.google.com. A +norecurse +tcp

- **Take a screenshot of the results of the final query for your lab notebook.**

```
evelynng@ada:~$ dig @ns2.google.com. console.cloud.google.com. A +norecurse +tcp

; <<>> DiG 9.20.24-1ubuntu0.3-Ubuntu <<>> @ns2.google.com. console.cloud.google.com
. A +norecurse +tcp
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 56557
;; flags: qr aa; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

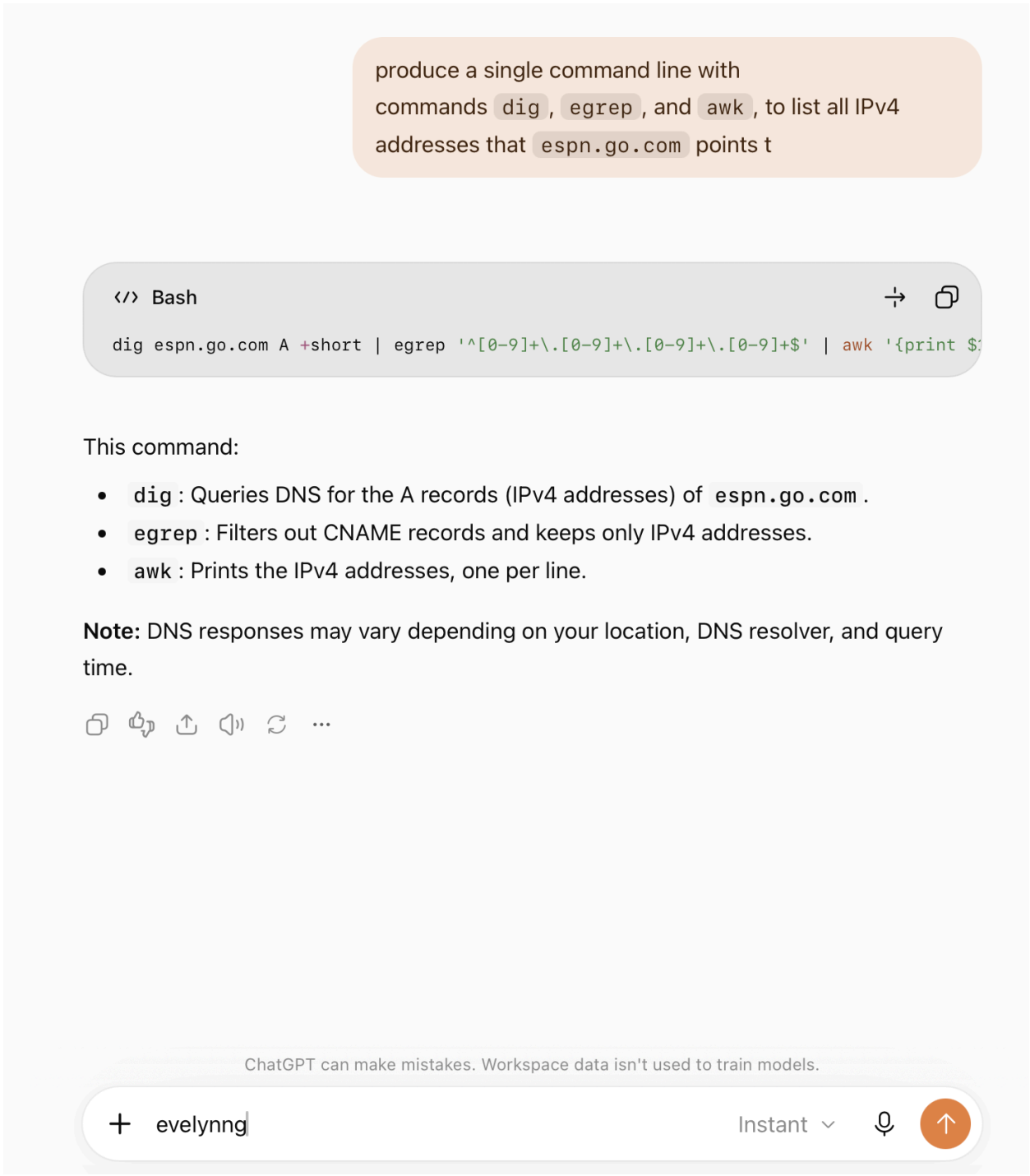
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;console.cloud.google.com.      IN      A

;; ANSWER SECTION:
console.cloud.google.com. 300   IN      CNAME   www3.l.google.com.
www3.l.google.com.      300     IN      A       142.251.34.206

;; Query time: 9 msec
;; SERVER: 216.239.34.10#53(ns2.google.com.) (TCP)
;; WHEN: Mon Oct 12 08:42:18 PDT 2026
;; MSG SIZE  rcvd: 90
```

3. Reverse DNS lookups

- **Take a screenshot of the prompt and the command produced**



- **Take a screenshot of its results for your lab notebook**

```
/dev/loop-control: character special (10/237)
evelynng@ada:~$ dig espn.go.com A +short | egrep '^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$'
 | awk '{print $1}'
143.204.1.2
143.204.1.101
143.204.1.62
143.204.1.118
evelynng@ada:~$
```

- **Take a screenshot of the command and its results for your lab notebook**

```
evelynng@ada:~$ for ip in $(dig espn.go.com A +short | egrep '^[0-9]+\.[0-9]+\.[0-9
]+\.[0-9]+$'); do dig -x "$ip" +short; done | egrep '[[:alpha:]]' | awk '{print $1}
'
server-143-204-1-62.hio52.r.cloudfront.net.
server-143-204-1-101.hio52.r.cloudfront.net.
server-143-204-1-118.hio52.r.cloudfront.net.
server-143-204-1-2.hio52.r.cloudfront.net.
```

4. Host enumeration

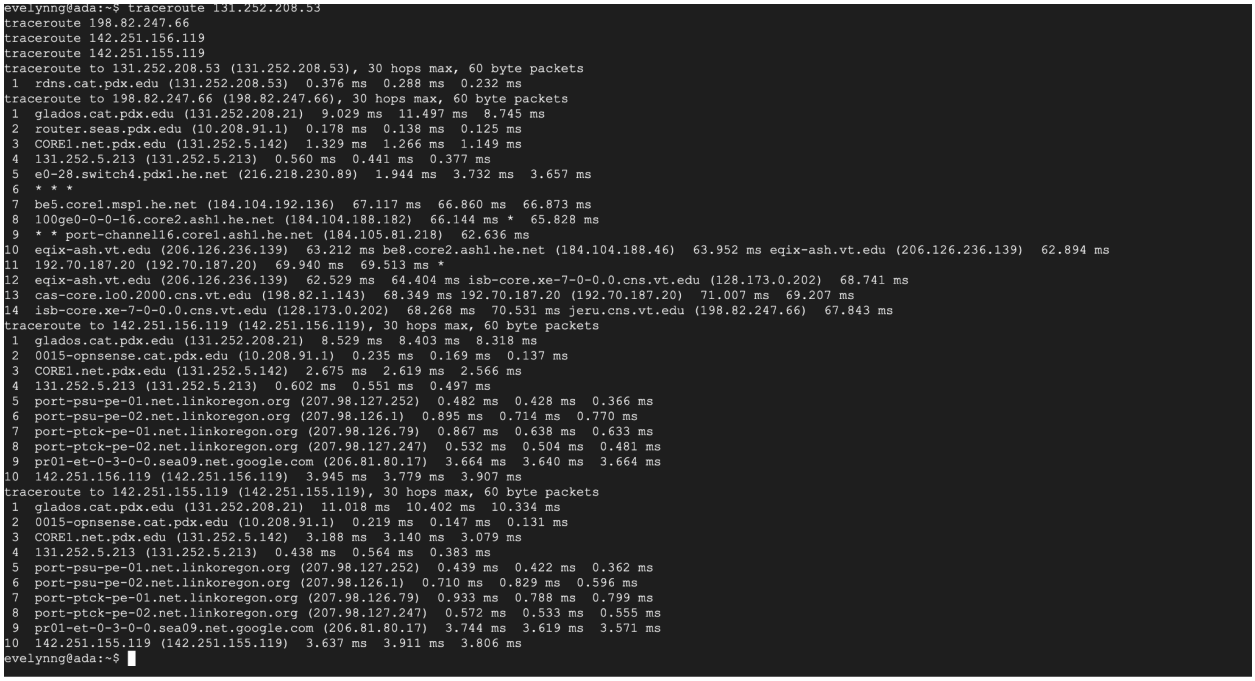
- **Take a screenshot of the results in your lab notebook**

```
evelynng@ada:~$ cat 220hosts.txt | head -185 | tail -30
acura.cs.pdx.edu.
astonmartin.cs.pdx.edu.
audi.cs.pdx.edu.
bentley.cs.pdx.edu.
bmw.cs.pdx.edu.
cadillac.cs.pdx.edu.
ferrari.cs.pdx.edu.
fiat.cs.pdx.edu.
ford.cs.pdx.edu.
honda.cs.pdx.edu.
hummer.cs.pdx.edu.
jaguar.cs.pdx.edu.
jeep.cs.pdx.edu.
lamborghini.cs.pdx.edu.
landrover.cs.pdx.edu.
lexus.cs.pdx.edu.
lotus.cs.pdx.edu.
maserati.cs.pdx.edu.
mazda.cs.pdx.edu.
mclaren.cs.pdx.edu.
mercedes.cs.pdx.edu.
nissan.cs.pdx.edu.
panoz.cs.pdx.edu.
porsche.cs.pdx.edu.
subaru.cs.pdx.edu.
toyota.cs.pdx.edu.
tvr.cs.pdx.edu.
ultima.cs.pdx.edu.
volvo.cs.pdx.edu.
vw.cs.pdx.edu.
evelynng@ada:~$
```

5. Geographic DNS #2

- **What geographic locations do ipinfo.io and DB-IP return?**
  - For DNS server **131.252.208.53**, DB-IP reports North Portland, Oregon, while ipinfo.io reports Hillsboro, Oregon.
  - For DNS server **198.82.247.66**, DB-IP reports Ashburn, Virginia, while ipinfo.io reports Blacksburg, Virginia.
- **Record one address for www.google.com from each result for your lab notebook.**
  - 142.251.156.119 for **131.252.208.53**
  - 142.251.155.119 for **198.82.247.66**

- **What are the geographic coordinates of each DNS server and the IP address it resolves for www.google.com?**
    - PSU DNS server (131.252.208.53): 45.5235, -122.6765
    - Virginia Tech DNS server (198.82.247.66): 37.2295, -80.4141
    - Google IP from PSU (142.251.156.119): 37.3861, -122.0839
    - Google IP from Virginia Tech (142.251.155.119): 37.3861, -122.0839
- **Take a screenshot of the results for your lab notebook.**



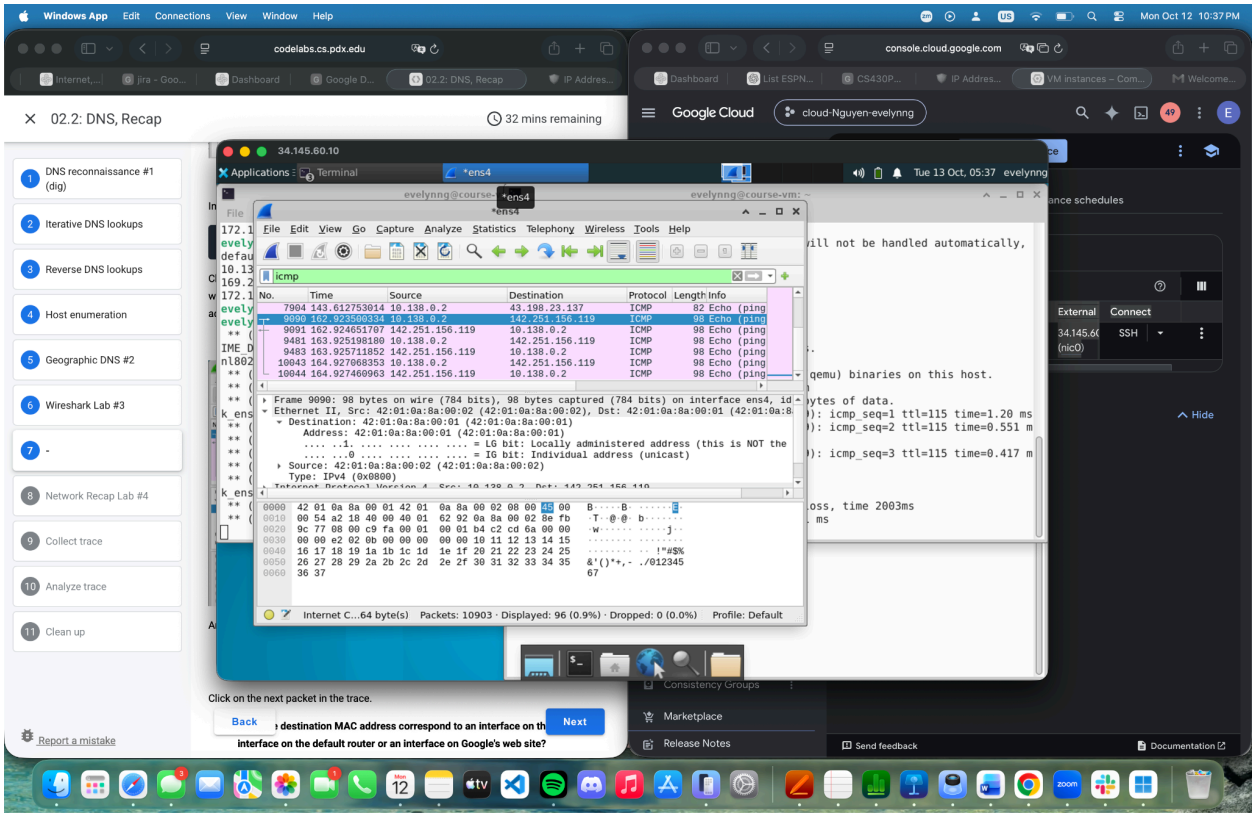
6. Wireshark Lab #3

In a terminal, using commands from prior labs, find the addresses and interfaces on the VM. Make a note of:

- The IP address of the VM: 10.138.0.2
- The name of the local virtual ethernet interface: ens4
- The IP address of the default router: 10.138.0.1

7. -

- **Take a screenshot of the bytes in the packet dump window as shown below**



- **Does the destination MAC address correspond to an interface on the VM, an interface on the default router or an interface on Google's web site?**
    - The destination MAC address corresponds to an interface on the default router because the VM sends the packet to the router to reach Google's website.
- **Does the destination MAC address correspond to an interface on the VM, an interface on the default router or an interface on Google's web site?**
    - The destination MAC address corresponds to an interface on the VM.

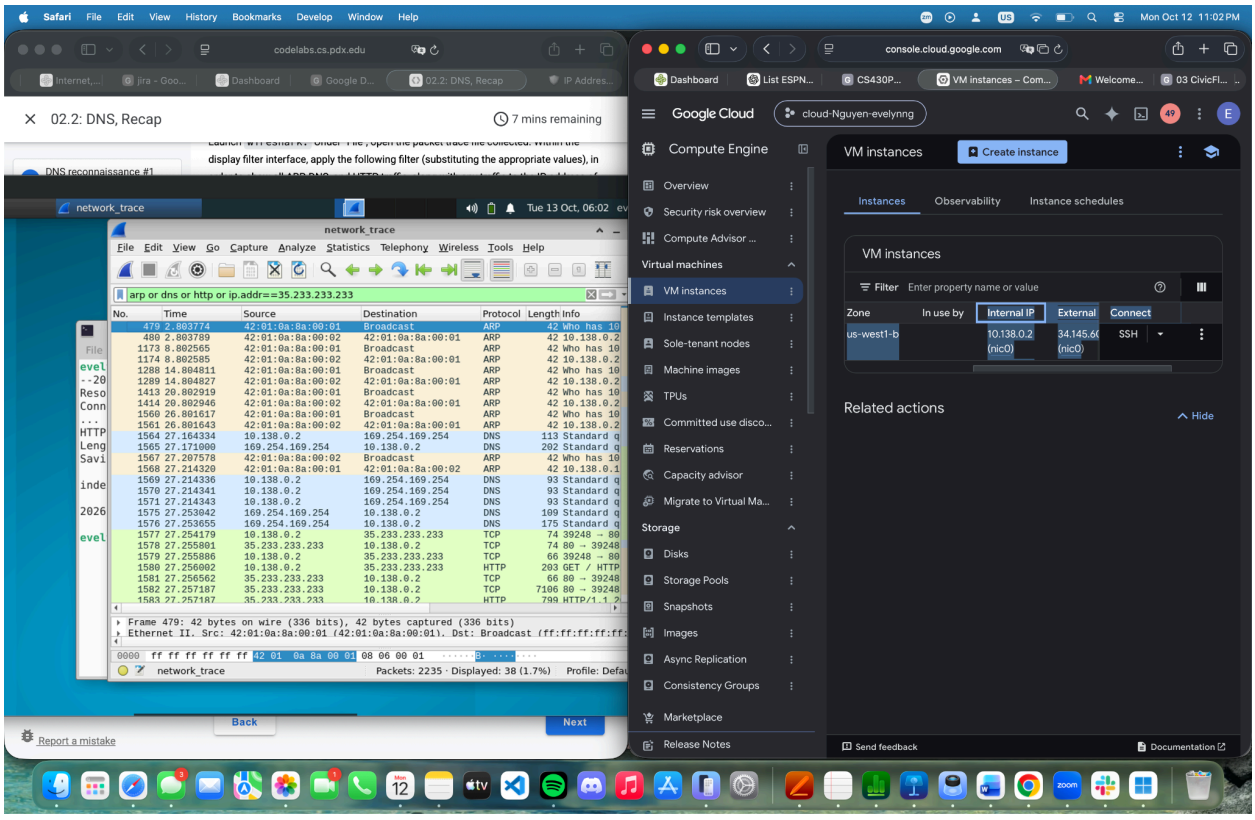
8. Network Recap Lab #4

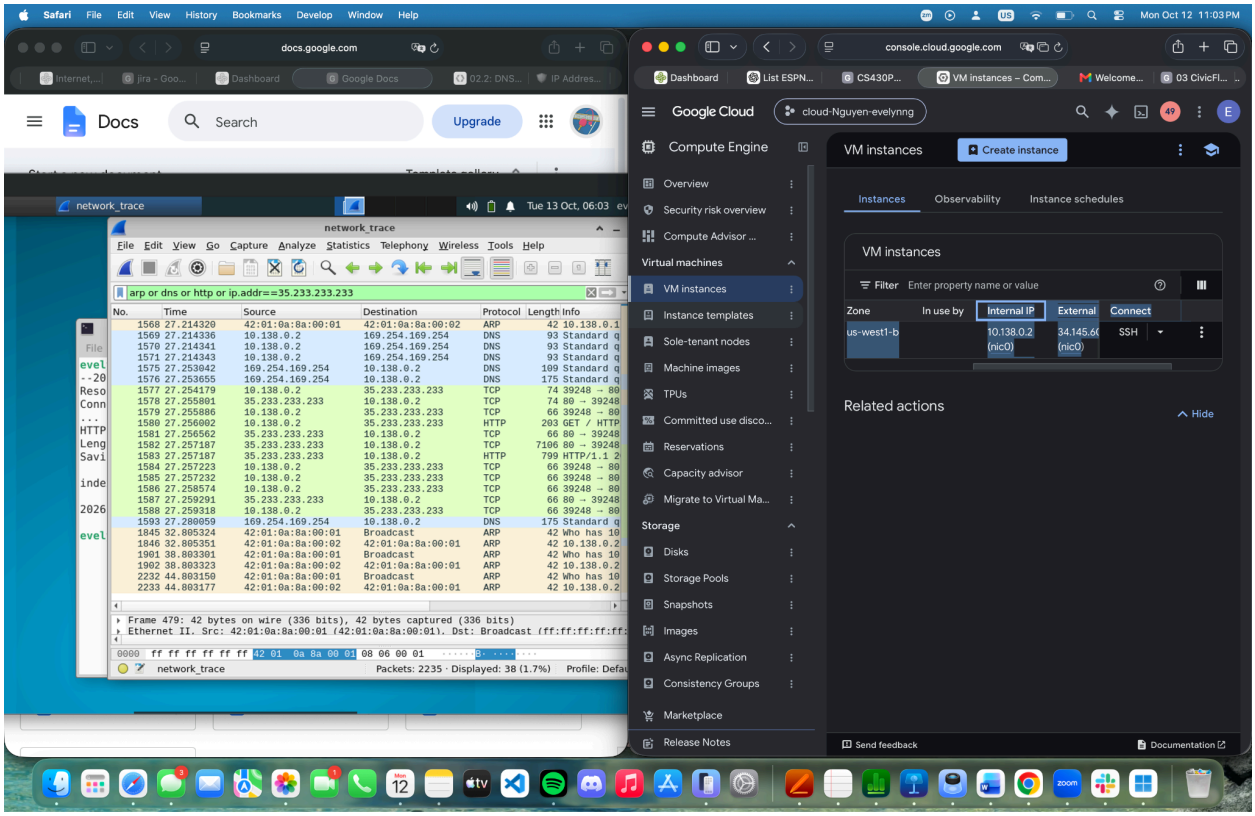
- **Find the IP address of <OdinId>.oregonctf.org, replacing <OdinId> with your OdinId:**

  -IP address of evelynng.oregonctf.org: 35.233.233.233

10. Analyze trace

- **Take a screenshot of the all of the packets returned within Wireshark that includes their packet numbers**





ARP

- **What packet numbers in the trace are the result of the VM attempting to get the hardware address of the default router?**
  - 479, 480
  - 1173, 1174
  - 1288, 1289
  - 1413, 1414
  - 1560, 1561
  - 1567, 1568
  - 1845, 1846
  - 1901, 1902
  - 2232, 2233
- **What is this hardware address?**
  - 42:01:0a:8a:00:01

DNS

- **What packet numbers in the trace correspond to the DNS request for the web site?**
    - 1569, 1570, 1571
    - 1575, 1576
    - 1593
- **What is the IP address of the local DNS server being queried?**

    - 169.254.169.254

<u>TCP</u>

- **What packet numbers in the trace correspond to the initial TCP handshake for the web request?**
    - 1577, 1578, 1579
- **How long does it take to perform the initial TCP handshake?**
    - Time of 1579 - Time of 1577 = 27.255886 - 27.254179 = 0.0017 07sec

HTTP

- **What packet numbers in the trace correspond to the actual HTTP request and response?**
    - 1580 and 1582
- **How long does it take to process the HTTP request after the handshake?**
    - Time of 1582 - Time of 1580 = 27.257187 - 27.256002 = 0.001185 sec